Quantum-inspired catalytic QUBO search for balanced hypergraph partitioning

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Abstract

Hypergraph partitioning assigns interacting vertices to balanced blocks while minimizing dependencies between them. Classical multilevel methods provide strong solutions. Quantum approaches offer alternative search dynamics, but faithful encoding of many-vertex costs alone does not ensure useful balanced search. Inspired by temporary catalyst interactions in quantum annealing, we adapt the principle of changing the search path while restoring the target energy at its endpoint to classical variational optimization. Incidence-dependent QUBO interactions control intermediate free-energy response without replacing the cut-net objective. A classical free-energy machine tests the construction on 30 original hypergraphs with one shared 48-replica policy. It improves 15 KaHyPar references at exact balance, but none at 1% or 3% imbalance. Preprocessing costs and numerical sensitivity limit practical gains. These results identify both the possibilities and limitations of quantum-inspired catalytic search.

Keywords: quantum-inspired optimization, QUBO, hypergraph partitioning, catalysts, free-energy machine

Introduction

Hypergraph partitioning allocates interacting tasks or components to balanced blocks while limiting shared resources. A net represents a signal or dependency involving many vertices; the cut-net objective charges it once when it crosses blocks. This model is used for circuit netlists and distributed computations[1,2]. Multilevel methods combine contraction and local refinement[3,4], and KaHyPar and its parallel successor provide strong classical references[5,6]. For an energy-based solver, the challenge is to preserve the many-vertex objective while creating useful motion toward a balanced partition. Encoding and dynamics answer different questions: what is minimized, and which states can be reached.

Quantum annealing searches through time-dependent Hamiltonians[7–9], while quantum approximate optimization algorithms (QAOA) use alternating parameterized circuits[10]. Ising formulations connect these methods to discrete optimization[11]. Quadratic unconstrained binary optimization (QUBO) uses two-body interactions. Exact quadratization represents many-body costs using auxiliary variables[12–14]. Graph partitioning has been tested on D-Wave hardware and

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through hybrid QUBO decomposition[15], while constraint-preserving mixers and drivers address fixed block sizes[16,17]. These constructions motivate hypergraph optimization, but an edge-cut result does not establish performance on the many-pin cut-net objective.

Recent hypergraph studies make this distinction explicit. Li et al. compare objective encodings and test simulated QAOA and annealing on small 3-uniform inputs[18]. Ibrahim et al. optimize distributions over partitions for fairness[19]; their target differs from cut-net minimization. These studies leave a search question unresolved: how can a faithful QUBO energy support effective balanced search on real hypergraphs? Preserving the target minimum does not determine the response along the search path. The representation and balance constraints also shape the intermediate dynamics, so endpoint fidelity alone is insufficient.

Temporary catalyst interactions offer a way to address the path while restoring the target energy at the endpoint. In quantum annealing, such interactions can alter transitions or gaps[20–22], with benefits depending on the model and schedule[23,24]. We take this path-design principle as the starting point for a classical variational search: a temporary energy term contributes its expectation under the trial distribution and vanishes before the final readout. This transfers the role of a catalyst rather than reproducing quantum evolution. For hypergraphs, incidence structure supplies both local couplings and extended balanced directions. The question is whether these directions can strengthen the intermediate response without replacing the objective. Overlap-gap theory provides a broader motivation: disconnected ranges of near-optimal overlaps can obstruct specified stable algorithms on specified random ensembles[25–27]. Related restrictions apply to shallow local quantum algorithms[28,29], and recent work compares quantum and mean-field-inspired searches[30]. These results do not establish an overlap gap for the real netlists studied here; our mechanism analysis concerns local response rather than a global escape theorem.

We realize this classical counterpart through temporary incidence-dependent QUBO interactions that preserve the cut-net endpoint while controlling balanced variational response. Exact examples identify objective distortion; a reduced free-energy Hessian connects the catalyst to local stability; and collective modes extend the available directions. A free-energy machine (FEM)[31] supplies a scalable classical testbed. On 30 original hypergraphs, one shared policy improves 15 KaHyPar references at exact balance, but none at 1% or 3% imbalance. The real-instance comparisons assess practical partition quality and computational cost. Preprocessing costs and numerical sensitivity constrain the practical gains.

Results

Preserving the cut-net objective

Two-way hypergraph partitioning divides a vertex set into two disjoint blocks, subject to a capacity bound, while minimizing the weight of nets that cross between them[5,18]. A graph joins pairs of vertices; a hypergraph $\mathcal{H} = (V, \mathcal{E})$ permits a net $e \in \mathcal{E}$ to contain any number of vertices. We write $n = |V|$, $m = |\mathcal{E}|$, $r_e = |e|$ and $w_e \geq 0$ for its weight. We first ask whether a binary energy preserves this partitioning objective.

Binary $x_i \in \{0, 1\}$ records the block of vertex i. For a graph with edge set E_2, the weighted edge-cut cost is

$$E_{\mathrm{G}}(\mathbf{x}) = \sum_{\{i,j\} \in E_2} w_{ij}(x_i - x_j)^2. \tag{1}$$

Each edge contributes its weight when its endpoints occupy different blocks. For a hypergraph, $k_e = \sum_{i \in e} x_i$ counts the vertices of net e assigned to one block. The cut-net objective is

$$E_{\mathrm{cut}}(\mathbf{x}) = \sum_{e} w_e \mathbb{I}(0 < k_e < r_e). \tag{2}$$

A crossing net is charged once, irrespective of the number of separated vertex pairs. For example, a four-vertex net split 1:3 or 2:2 costs one net weight in either case; a pair surrogate instead counts three or four crossing pairs. For two-vertex nets, cut-net and graph cut coincide.

The unit-load two-way problem also imposes balance: exact balance has $N = \sum_i x_i = b = \lfloor n/2 \rfloor$, whereas an allowed imbalance band specifies integer bounds on N. All reported cuts are scored on the original nets and accompanied by their actual imbalance. An allowed imbalance is an upper bound; a more balanced output remains feasible at that setting.

Quadratic unconstrained binary optimization (QUBO) expresses an energy as $\mathbf{z}^{\mathsf{T}}\mathbf{Q}\mathbf{z}+c_0$, where $\mathbf{z}$ is binary[11,12]. A many-vertex cut-net cost can be represented exactly by quadratic interactions and auxiliary variables[13]. Here two binary auxiliaries per net give an exact QUBO after auxiliary minimization; a quadratic penalty enforces balance (Methods). This construction fixes the target of the search. We next distinguish preserving that target from providing useful motion toward it.

The first obstacle is objective distortion. We compare two quadratic clique surrogates that sum $w_e k_e(r_e - k_e)$ or divide that quantity by $r_e - 1$. Both count pairs separated within each net. Even normalizing the pairs cannot make every crossing configuration of an arbitrary net cost exactly one. The difference affects the optimum, not just the quality of a heuristic.

Exact enumeration gives concrete counterexamples (Fig. 1). Each has 12 vertices, ten mixed-size nets and 924 balanced assignments. In example A, the native optimum is 7, but every minimum of the unit-pair surrogate has native cut 8. In example B, the native optimum is 5; the normalized surrogate selects native cut 6, while the unit-pair surrogate has minimum-state cuts ranging from 8 to 10. These are illustrative counterexamples, not estimates of average surrogate performance. The auxiliary-minimized QUBO recovers the native minima in both examples.

Table 1 separates surrogate distortion from exact quadratic encoding. Preserving the objective still leaves a search barrier. Its exact-balance penalty creates a coordinate barrier: every balanced, auxiliary-minimized binary partition is a one-bit local minimum when $\lambda > D_{\max}$, where $D_{\max}$ is the maximum weighted incidence degree, regardless of its cut. Faithfulness alone therefore cannot supply a useful search dynamic (Methods).

Incidence controls balanced variational response

The encoding barriers motivate a temporary structural response while the vertex labels remain uncertain. We keep the complete exact QUBO and change its intermediate path. A pair term $\gamma(t)C_{\text{pair}}$ redistributes interactions within nets, while an auxiliary term favors polarized auxiliary states:

$$\begin{aligned} C_{\text{pair}} &= \sum_e w_e a_e \sum_{i<j\in e} z_i z_j (x_i - x_j)^2, \\ C_{\text{aux}}(t) &= -\sum_e \kappa_e(t)(u_e - v_e)^2. \end{aligned} \tag{3}$$

Here u_e, v_e are the two binary auxiliaries for net e. The scales z_i depend on incidence degree and local degree contrast; a_e preserves total coupling mass when these scales change. Both catalysts vanish at the endpoints. Intermediate per-spin temperatures supply another variational control, with their offsets also vanishing before the final original-QUBO phase (Fig. 2). Consequently the final binary objective is unchanged, and a penalty $\lambda > D_{\max}$ guarantees exactly balanced global minima (Methods). This guarantee concerns the encoding, not the output of a finite heuristic.

To identify the response controlled by these interactions, let $p_i = \Pr(x_i = 1)$ and examine the reduced free energy after minimizing each four-state auxiliary block. For even n, at uniform temperature and the uncertain state $p_i = 1/2$, the Hessian is

$$\begin{aligned} \mathbf{H} = {} & 4T\mathbf{I} + 2\lambda(\mathbf{1}\mathbf{1}^{\mathsf{T}} - \mathbf{I}) \\ & - \mathbf{B}^{\mathsf{T}} \operatorname{diag}(w_e^2 \chi_e / T)\mathbf{B} \\ & - 2\gamma \mathbf{A}_{\text{off}}, \end{aligned} \tag{4}$$

where $\mathbf{B}$ is incidence, χ_e is the auxiliary susceptibility and $\mathbf{A}_{\text{off}}$ is the graph-dependent off-diagonal pair operator. Projection onto the zero-mean subspace isolates balanced fluctuations. Entropy stabilizes uncertainty; auxiliary susceptibility and catalytic geometry shift the unstable directions. Degree alone cannot determine that operator. The exact curvature expression agrees with automatic second derivatives on weighted test instances. Its minimum eigenvalue changes systematically with catalyst strength on the illustrated hypergraph (Fig. 2(c)).

Local incidence weights need not supply an extended direction across a large graph. We therefore also use zero-mean, unit-norm low-frequency modes $\mathbf{v}_\ell$ of incidence-derived Laplacians, with both unscaled and degree-normalized geometry. Spectral partitioning supplies the classical structural motivation[32,33]. The modes are computed from the original incidence. Their collective

Table 1 Cut-net encodings

Construction	Order	Binary variables	Property relevant to cut-net search
Unit-pair clique	2	n	Different objective; example A selects cut 8 instead of 7.
Normalized clique	2	n	Different objective; example B selects cut 6 instead of 5.
Complete auxiliary QUBO	2	$n + 2m$	Exact cost after auxiliary minimization; extra spins and couplings.

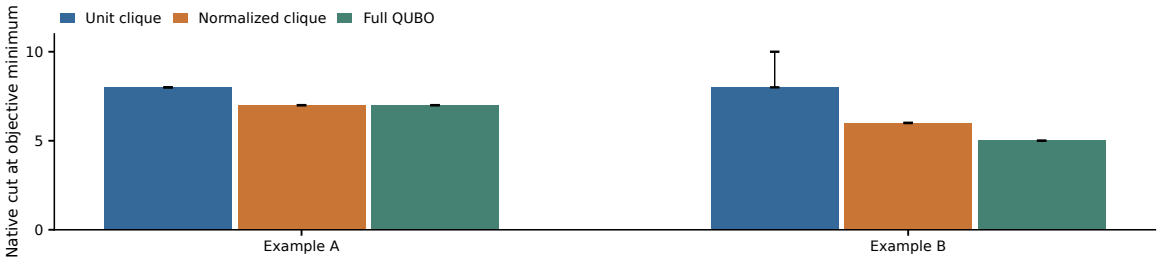


Figure 1 Preserving the cut-net optimum; Native cut among global minima of each objective on two exact counterexamples; Bar heights use the best native cut among tied minima; the whisker on example B's unit clique spans all native cuts among its minima, not an uncertainty interval; The complete auxiliary QUBO recovers the native optima after auxiliary minimization

quadratic catalyst is

$$C_{\mathrm{col}}(t) = -\frac{\rho(t)}{2} \sum_{\ell} \theta_\ell \left[(\mathbf{v}_\ell^{\mathsf{T}} \mathbf{s})^2 - \|\mathbf{v}_\ell\|^2 \right], \qquad s_i = 2x_i - 1. \tag{5}$$

with $\theta_\ell \geq 0$ and $\sum_\ell \theta_\ell = 1$. For one mode, its expected-energy Hessian adds $-4\rho(\mathbf{v}\mathbf{v}^{\mathsf{T}} - \mathrm{diag}(v_i^2))$. The mean-balance rank-one term annihilates $\mathbf{v}$, while this catalyst changes curvature along $\mathbf{v}$ by $-4\rho(1 - \sum_i v_i^4)$. Thus an extended mode supplies a strong balanced instability; a mode localized on one vertex supplies none. This statement concerns local variational curvature, not a quantum gap.

Vertices incident only to singleton nets have zero cut response, but still enter balance. We retain these spins and add a temporary quadratic coupling on a zero-sum random signed direction supported on them. This null-support catalyst encourages their balanced polarization without supplying a binary partition or repairing any vertex. Both collective coefficients vanish at the endpoints. A shared replica portfolio combines individual modes, mode subspaces, local catalysts and per-spin annealing. Component contrasts handle disconnected cut-active supports (Methods).

These calculations establish a local mechanism: the catalyst changes balanced variational curvature while preserving the final energy. The schematic overlap-gap and trapping panels in Fig. 2(d)–(f) illustrate the distinction from global solution geometry. An overlap gap does not require a flat surface or vanishing gradients, and neither its existence nor its removal follows from this Hessian. The next test is therefore empirical: whether the available directions produce useful partitions under a fixed search policy. A computed variational slice is provided in Supplementary Note 3. We next evaluate these classical search directions under one fixed policy on the original hypergraphs.

Solving catalytic QUBOs under capacity constraints

The local response analysis suggests search directions; their usefulness requires a solver test. We use a free-energy machine (FEM), a classical variational annealing method[31]. Each replica

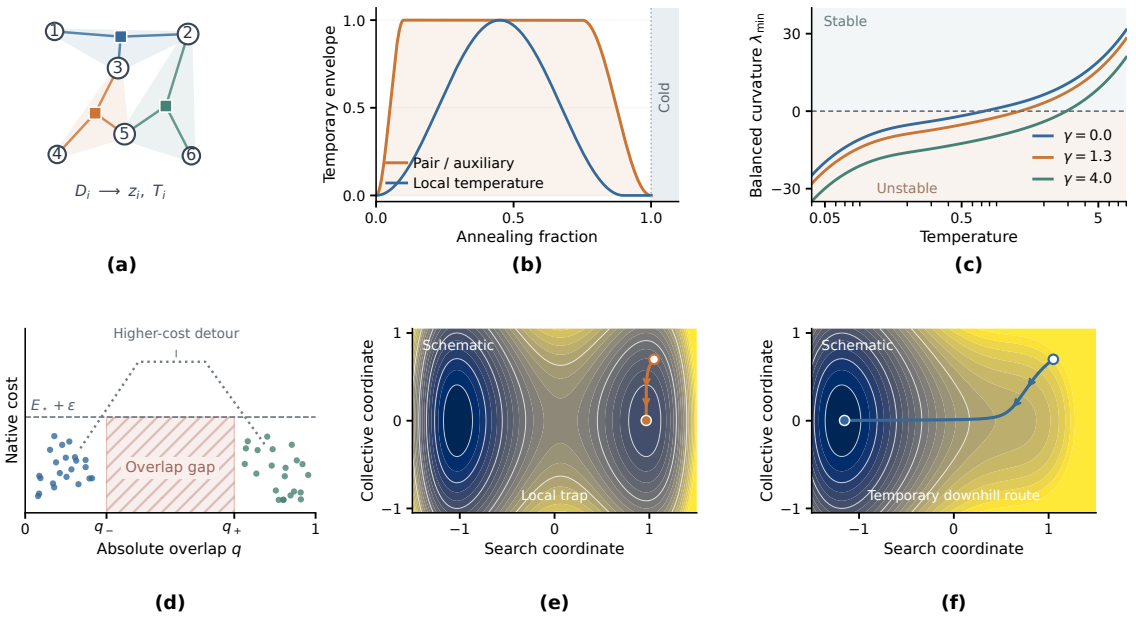


Figure 2 Catalyst landscapes; (a) A schematic shared-net hypergraph; squares represent nets and circles represent vertices, with larger circles indicating higher degree; Degree and local incidence contrast determine structural scales and temperature offsets; **(b)** Temporary pair/auxiliary envelopes and local-temperature offsets vanish before the cold phase; **(c)** Minimum balanced-subspace curvature from Equation (4) on example B, with penalty 0.01 and pair strengths 0, 1.3 and 4; Zero marks the local stability boundary; The curves are classical free-energy curvatures, not quantum spectral gaps; **(d)** Schematic OGP: near-optimal pairs have absolute overlap outside (q_-, q_+); the dotted detour leaves the near-optimal set; **(e)** An illustrative two-basin potential traps gradient descent in a local minimum; **(f)** A temporary term changes the descent direction and removes that local trap in the illustration; Contours indicate increasing potential; open circles mark the common initial state and filled circles the endpoints; Panels (d)–(f) are schematic, not measurements on benchmark graphs; The catalyst vanishes at the end of a search, restoring the original landscape; no OGP escape or measured barrier reduction is inferred

evolves vertex probabilities and four-state net-auxiliary distributions by balancing expected energy against entropy. Cooling produces binary candidates, which are evaluated on the original cut nets and capacity constraints (Methods). This separates the proposed energy path from the algorithm used to explore it.

KaHyPar supplies the classical reference. This multilevel hypergraph partitioner contracts vertices, constructs an initial partition and refines it as the contractions are undone[4,5]. Its partitions are evaluated independently and do not initialize FEM; Supplementary Note 1 describes the reference configurations.

We test FEM on 30 original hypergraphs with unit vertex loads and net weights at allowed imbalance 0%, 1% and 3%, with a separate 48-replica batch at each setting. One shared policy combines local, unscaled and degree-normalized groups with matched nearby random fields and 12,144 updates. Every vertex and net auxiliary remains in the QUBO; the bands add bounded slack (Methods). No classical partition initializes the search. These primary measurements use FP64. Table 2 pairs native cut with measured imbalance (%) at matching KaHyPar capacity settings. Bold marks the lower cut or charged time within each pair, including ties; the recorded references are not asserted optima.

The outcome depends strongly on capacity. FEM improves 15 of 30 recorded references at exact balance, but none at 1% or 3%; each band has four ties. Large gaps remain, including strict cuts of 2,544 against 1,204 on *IBM13* and 318 against 68 on *scircuit*. The expanded feasible set also does not guarantee a better finite search: *IBM01* changes from 325 at 0% to 335 at 3%, and *cfd1* from 2,064 to 2,250. The three settings are independent searches with equal trajectory counts, so strict results are not inserted

Table 2 QUBO cut and measured imbalance (%);

	Allowed imbalance 0%		Allowed imbalance 1%		Allowed imbalance 3%	
Graph	FEM	KaHyPar	FEM	KaHyPar	FEM	KaHyPar
IBM01	325 (0.000)	**287** (0.000)	368 (0.988)	**220** (0.471)	335 (2.964)	**202** (2.760)
IBM02	477 (0.000)	**367** (0.000)	454 (0.990)	**350** (0.520)	420 (3.000)	**337** (2.989)
IBM03	**1,171** (0.000)	1,247 (0.000)	1,160 (0.994)	**962** (0.000)	1,143 (2.939)	**957** (0.199)
IBM04	**621** (0.000)	696 (0.000)	636 (0.996)	**598** (0.996)	609 (2.995)	**582** (1.898)
IBM05	1,901 (0.000)	**1,807** (0.000)	1,896 (0.402)	**1,725** (0.000)	1,895 (0.450)	**1,719** (1.036)
IBM06	**1,080** (0.000)	1,335 (0.000)	1,078 (0.560)	**1,023** (0.923)	1,080 (0.892)	**988** (1.994)
IBM07	**1,163** (0.000)	1,428 (0.000)	1,144 (0.997)	**948** (0.997)	1,206 (2.996)	**910** (2.992)
IBM08	1,238 (0.000)	**1,200** (0.000)	1,181 (0.998)	**1,140** (0.000)	1,211 (2.276)	**1,140** (0.000)
IBM09	756 (0.000)	**703** (0.000)	723 (0.996)	**627** (0.974)	685 (2.996)	**620** (2.663)
IBM10	2,070 (0.000)	**1,626** (0.000)	1,732 (0.746)	**1,340** (0.000)	1,720 (0.668)	**1,333** (0.349)
IBM11	**1,581** (0.000)	1,671 (0.000)	1,458 (0.998)	**1,079** (0.989)	1,417 (2.999)	**1,062** (2.891)
IBM12	**2,443** (0.000)	3,138 (0.000)	2,457 (0.788)	**1,996** (0.979)	2,492 (2.895)	**1,936** (2.476)
IBM13	2,544 (0.000)	**1,204** (0.000)	2,508 (1.000)	**848** (0.404)	1,921 (3.000)	**848** (0.404)
IBM14	3,259 (0.000)	**2,273** (0.000)	3,088 (1.000)	**1,849** (0.000)	3,577 (3.000)	**1,849** (0.000)
IBM15	4,573 (0.000)	**4,554** (0.000)	4,721 (0.999)	**2,839** (0.722)	4,145 (0.586)	**2,741** (1.868)
IBM16	**2,907** (0.000)	4,044 (0.000)	3,770 (1.000)	**2,117** (0.832)	3,544 (2.746)	**1,969** (2.941)
IBM17	3,446 (0.000)	**3,390** (0.000)	3,366 (0.999)	**2,351** (0.965)	2,784 (3.000)	**2,310** (2.600)
IBM18	**2,197** (0.000)	2,772 (0.000)	2,231 (0.537)	**1,961** (0.000)	2,326 (2.989)	**1,915** (1.355)
2cubes_sphere	**4,125** (0.000)	6,892 (0.000)	3,991 (0.989)	**3,922** (0.993)	3,951 (2.999)	**3,884** (2.993)
G2_circuit	**624** (0.000)	1,144 (0.000)	**624** (0.779)	**624** (0.000)	**624** (0.779)	**624** (0.000)
PGPgiantcompo	463 (0.000)	**396** (0.000)	467 (0.449)	**352** (0.000)	457 (2.959)	**350** (0.000)
airfoil_2d	375 (0.000)	**372** (0.000)	373 (0.999)	**328** (0.000)	342 (2.997)	**324** (2.899)
bcsstk29	372 (0.000)	**360** (0.000)	**360** (0.172)	**360** (0.000)	**360** (0.172)	**360** (0.000)
ca-CondMat	4,473 (0.000)	**4,202** (0.000)	4,406 (0.994)	**3,918** (0.916)	4,215 (3.000)	**3,855** (3.000)
cfd1	**2,064** (0.000)	3,121 (0.000)	2,174 (0.999)	**1,564** (0.725)	2,250 (2.998)	**1,564** (0.725)
finan512	**146** (0.000)	169 (0.000)	**146** (0.000)	**146** (0.000)	**146** (0.000)	**146** (0.000)
parabolic_fem	**1,026** (0.000)	1,256 (0.000)	**1,026** (0.013)	**1,026** (0.000)	**1,026** (0.013)	**1,026** (0.000)
scircuit	318 (0.000)	**68** (0.000)	277 (0.999)	**46** (0.513)	46 (2.451)	**42** (2.273)
shallow_water2	**768** (0.000)	909 (0.000)	768 (0.764)	**640** (0.000)	768 (0.601)	**640** (0.000)
thermal1	**255** (0.000)	291 (0.000)	257 (0.102)	**239** (0.111)	257 (0.099)	**239** (0.114)

into the band columns. These contrasts locate the current policy's usefulness at selected exact-balance settings, rather than establishing general competitiveness.

The exact-balance policy retains cuts 146 on *finan512*, 255 on *thermal1*, 624 on *G2_circuit* and 1,026 on *parabolic_fem*. KaHyPar's 3% runs on *finan512*, *G2_circuit* and *parabolic_fem* also return exactly balanced partitions, with cuts 146, 624 and 1,026. These remain feasible classical ties even when the separately configured 0% references are worse. Classical rounding of eight continuous incidence modes matches these three cuts and reaches 272 on *thermal1*; FEM improves that last cut by 17 nets. Spectral information alone explains several strong outcomes, and its rounded partitions are assessed separately.

The FEM score includes the specified original-QUBO one-bit descent, whose time is charged for all replicas. On 29 exact-balance instances, the winning raw partition is unchanged. On *scircuit*, the best feasible raw cut is 373; a different raw replica with count residual 2 reaches cut 318 through two vertex and four auxiliary flips. This exception is reported explicitly. Raw feasibility counts, cut differences and winner flip counts for all three balance settings are in Supplementary Note 4.

Twenty inputs informed earlier studies; ten extend the fixed policy to the full *IBM01–IBM18* family and three further matrix-derived hypergraphs. One base seed per graph does not establish seed robustness or fully unseen generalization. Group comparisons and mechanism controls are retained in Supplementary Notes 4 and 5.

Preprocessing and precision limit practical gains

Solution quality must be assessed together with the cost of obtaining it. We distinguish the primary pipeline measurements from kernel-fusion and precision controls, because their numerical paths and preprocessing differ.

Table 3 reports the cost of the primary FP64 policy. It charges both mode constructions, graph setup, compilation/capture, all 48 trajectories, checkpoints and all-replica descent. CPU construction costs are summed even when jobs overlap. On *parabolic_fem*, the original CPU modes cost 1466.81 s and the strict GPU loop 467.52 s; the preprocessing dominates its 1971.66 s charged total. On *scircuit*, both an unsuccessful CPU mode attempt and its certified A100 replacement are charged. The longest entries therefore measure the complete historical pipeline, including costly preprocessing, rather than GPU arithmetic alone.

A separate precision control on *IBM01* reuses certified mode arrays across three capacity settings (Supplementary Note 6). FP32 reduces search from 19.73–20.09 s to 14.32–14.60 s, while charged totals fall from 27.01–27.82 s to 21.66–22.39 s. The smaller pipeline gain reflects costs outside the accelerated search. Precision also changes the winning trajectory: strict QUBO cut worsens from 325 to 438, whereas 3% QUBO improves from 314 to 284. The cut changes have opposite signs, so these three comparisons on one input do not establish that FP32 generally improves or worsens quality. The measurements were made in separate runs; a broader paired comparison is required to assess the reproducibility of quality and timing changes.

This separates two practical limitations. Mode construction can dominate the historical pipeline, while repeated count-root and sparse actions still cost time inside the search. Component profiling and kernel-fusion controls identify both targets (Supplementary Note 6); they are separate protocols, not revised primary timings. Hardware and reference budgets also differ, so the recorded costs do not establish an equal-budget or equal-hardware speedup.

Discussion

The central result is a connection between objective fidelity and controllable search response. Clique substitution can change the desired optimum; exact encoding removes that error but does not remove the variational and coordinate barriers. Incidence-dependent temporary interactions supply local and extended balanced directions while the final cut-net energy is restored. The Hessian explains this mechanism within the stated trial distribution. The quantum-inspired ingredient is the temporary modification of the search energy; the resulting algorithm is a classical optimization of variational free energy.

The practical evidence is narrower than the mechanism. The shared policy improves selected exact-balance references but loses its comparative gains when capacity is relaxed. Spectral rounding and exactly balanced KaHyPar outputs already match the strong cuts on *finan512*, *G2_circuit* and *parabolic_fem*; the collective modes therefore carry useful classical information that must be included in any interpretation of the gains. Large remaining quality gaps and costly preprocessing limit the present solver. The single-input precision control raises a numerical question; it does not establish general FP32 degradation. FP32 addresses part of the computation, but broader paired quality and cost measurements are needed before revising benchmark claims.

An overlap-gap claim would require a specified ensemble, threshold, algorithm class and scaling analysis. Local curvature and a few returned partitions cannot determine forbidden overlaps throughout the low-cost set. Correlated variational models, including cluster mean-field[34], autoregressive distributions[35] and tensor networks[36], could test which barriers arise from independent vertex marginals. Transverse catalysts require separate quantum experiments. The current framework makes these questions testable by keeping the objective, intermediate response and endpoint evaluation explicit.

Methods

We develop a classical variational search from the idea of temporary catalyst interactions in quantum annealing. The derivation first fixes an exact

Table 3 QUBO charged time (s);

	Allowed imbalance 0%		Allowed imbalance 1%		Allowed imbalance 3%	
Graph	FEM	KaHyPar	FEM	KaHyPar	FEM	KaHyPar
IBM01	**25.35**	128.68	**20.73**	64.32	**21.98**	64.67
IBM02	**31.42**	128.30	**27.52**	64.61	**29.46**	64.44
IBM03	**33.19**	128.61	**29.23**	64.15	**29.79**	64.51
IBM04	**34.91**	128.67	**31.20**	64.23	**33.00**	65.10
IBM05	**35.75**	128.50	**32.00**	64.26	**33.43**	64.30
IBM06	**37.31**	129.22	**34.35**	64.66	**35.32**	66.49
IBM07	**49.79**	128.73	**48.05**	64.40	**49.88**	65.25
IBM08	**105.68**	128.37	103.51	**64.82**	105.71	**64.52**
IBM09	**59.74**	128.67	**58.02**	64.48	**59.28**	64.66
IBM10	**84.20**	129.23	82.91	**64.43**	85.31	**64.85**
IBM11	**78.65**	128.55	77.93	**64.41**	80.82	**64.90**
IBM12	**91.24**	128.81	90.88	**66.19**	91.72	**66.35**
IBM13	**103.27**	130.33	103.87	**65.48**	106.56	**64.40**
IBM14	212.18	**128.97**	219.53	**66.70**	220.02	**65.71**
IBM15	208.97	**65.63**	214.57	**64.59**	216.81	**69.89**
IBM16	297.77	**136.65**	304.09	**68.51**	307.32	**72.02**
IBM17	215.52	**144.28**	221.60	**66.23**	223.75	**65.19**
IBM18	252.76	**130.88**	260.83	**67.02**	263.85	**67.20**
2cubes_sphere	132.94	**129.14**	134.23	**67.97**	**134.80**	155.80
G2_circuit	163.52	**129.09**	170.21	**64.91**	173.42	**67.27**
PGPgiantcompo	**24.87**	128.57	**20.57**	64.31	**21.61**	64.40
airfoil_2d	**28.33**	128.35	**24.24**	64.16	**24.81**	64.23
bcsstk29	**32.47**	129.15	**28.11**	64.27	**29.45**	64.41
ca-CondMat	**40.99**	128.85	**36.71**	65.58	**37.52**	66.07
cfd1	**94.29**	130.68	93.73	**64.89**	95.12	**64.73**
finan512	**86.46**	128.59	86.11	**64.52**	88.30	**64.52**
parabolic_fem	1971.66	**133.58**	2004.51	**67.59**	2007.91	**80.87**
scircuit	3352.19	**132.23**	3419.80	**65.11**	3744.58	**66.07**
shallow_water2	**74.41**	128.99	73.75	**64.73**	75.30	**64.85**
thermal1	**106.93**	128.47	105.68	**64.90**	107.03	**66.02**

cut-net QUBO, then specifies the probability distribution and balance update, and finally constructs interactions that change the intermediate free energy while vanishing at readout. The evaluation protocol connects these choices to partition quality and computational cost.

Problem definition and exact endpoint

For nonnegative net weights, the objective is $E_{\mathrm{cut}}(\mathbf{x}) = \sum_e w_e \mathbb{I}(0 < k_e < r_e)$ with $\sum_i x_i = b$. For odd n, block sizes are b and $n - b$. Two auxiliaries per nonempty net give

$$\min_{u_e, v_e \in \{0,1\}} \big[1 + (r_e - 1 - k_e)u_e + (k_e - 1)v_e\big] = \mathbb{I}(0 < k_e < r_e). \tag{6}$$

The complete QUBO multiplies each net term by w_e and adds the balance penalty.

Let $D_i = \sum_{e \ni i} w_e$ and $D_{\max} = \max_i D_i$ denote weighted incidence degrees. Equation (6) follows by examining the three cases $k_e = 0$, $0 < k_e < r_e$ and $k_e = r_e$. Minimizing the two auxiliary bits gives costs 0, 1 and 0, respectively. A state with imbalance $d = |\sum_i x_i - b|$ can be balanced by flipping d vertices. Its native cut can increase by at most $dD_{\max}$, whereas removing its penalty saves λd^2. Thus $\lambda > D_{\max}$ excludes imbalanced global minima. We use $D_{\max} + 1$ and retain the complete $n + 2m$ binary state for independent scoring.

This equality concerns binary energies after auxiliary minimization; it does not identify the variational QUBO energy with the expected

native cut. At a fixed vertex distribution, the expectation of the auxiliary minimum can be smaller than the minimum expected energy of auxiliaries independent of vertices. Conditional auxiliaries that select a minimizing state for each vertex assignment recover the expected native cut. Our factorized vertex distribution and within-net auxiliary factors do not impose these vertex–auxiliary correlations. Auxiliary entropy therefore changes the finite-temperature variational landscape despite the exact binary endpoint.

For an allowed imbalance ϵ, let $c = \lceil n/2 \rceil$, $U = \lfloor (1+\epsilon)c \rfloor$, $L = n - U$ and $N = \sum_i x_i$. A bounded binary slack $S = \sum_j a_j b_j$ spans all integers from 0 to $U - L$. Its weights are powers of two followed by the remaining range. The matched 1% and 3% QUBOs replace exact balance by

$$E_{\mathrm{bal}} = \lambda (N - L - S)^2. \tag{7}$$

Minimizing slack at fixed vertices gives $\lambda \operatorname{dist}(N, [L, U])^2$. The same vertex-flip bound proves that $\lambda > D_{\max}$ excludes infeasible global minima. This uses $\lceil \log_2(U - L + 1) \rceil$ extra bits, while retaining every vertex and net auxiliary. No binary vertex is projected or repaired by a balance rule. Reported imbalance is $100[\max(n_0, n_1)/\lceil n/2 \rceil - 1]$; the smallest possible odd split therefore has zero reported imbalance.

The same coefficient implies an endpoint coordinate barrier. At a balanced state, flipping vertex i while fixing auxiliaries changes energy by

$$\Delta_i E = \lambda + (1 - 2x_i) \sum_{e \ni i} w_e (v_e - u_e) \geq \lambda - D_i > 0. \tag{8}$$

If auxiliaries are conditionally minimized, none of their single flips lowers energy either. Every such balanced state is therefore a one-bit local minimum, including poor cuts. One-bit descent can restore feasibility and auxiliary consistency but cannot improve these already feasible endpoints. This lemma is independently checked by exhaustive weighted small-instance enumeration and on 384 saved *finan512/thermal1* states. It concerns coordinate dynamics of the encoding, not OGP or a limitation on correlated continuous or quantum dynamics. The band encoding has the same one-bit barrier when $N = L + S$: vertex flips pay at least $\lambda - D_i$, and a single slack flip pays λa_j^2. Allowing more feasible block sizes does not remove this encoding-induced barrier.

Objective-fidelity checks

For clique objectives, we evaluate both unit pair weights and weights $1/\max(r_e - 1, 1)$. Exact enumeration scores every balanced assignment on 12 vertices. Fixed generation seeds 200–1199 define ten mixed-size nets; the first strict counterexample for each surrogate is retained, giving seeds 200 and 207 after eight examined graphs. All surrogate minimizers are included and the best native cut among ties is reported. These are explicitly selected mechanism examples, not a representative benchmark. All 924 assignments per example agree with the independently scored auxiliary QUBO after conditional auxiliary minimization. KaHyPar uses one ordinary seed-0 strict-balance run per example.

Variational distribution and update

The quantum catalyst motivates a temporary change to the search energy. In the classical variational formulation, this change enters through $\mathbb{E}_q C_t$, the expectation of the temporary interaction under a trial distribution q. No quantum state is evolved. The binary endpoint does not determine q, so we specify it before defining the updates.

Mean-field annealing replaces discrete states by continuous marginals[37]. FEM follows this variational approach[31], with entropy and expected energy defined for an explicit trial distribution[38]. For replica a, the distribution is $q_a(x, u, v) = \prod_i \mathrm{Bern}(x_i; p_i^a) \prod_e q_e^a(u_e, v_e)$. The free-energy objective contains the expected QUBO and catalysts, vertex negative entropy $\sum_i T_i(t)[p_i \ln p_i + (1 - p_i) \ln(1 - p_i)]$, and auxiliary negative entropy $T(t) \sum_{e,u,v} q_e(u, v) \ln q_e(u, v)$. The four-state auxiliary block is minimized exactly at each vertex update. It retains within-edge auxiliary correlation but no vertex–vertex correlation. For $\bar{k}_e = \sum_{i \in e} p_i$, its nonconstant state energies are 0, $w_e(r_e - 1 - \bar{k}_e) - \kappa_e$, $w_e(\bar{k}_e - 1) - \kappa_e$, and $w_e(r_e - 2)$. The reduced block contribution is $w_e - T \ln \sum_{u,v} \exp[-E_e^{\mathrm{nc}}(u, v)/T]$, where E_e^{nc} denotes these four nonconstant energies. Supplementary Note 3 specifies the projected variational slice and the illustrative landscapes used in

Fig. 2; the latter are schematic potentials, rather than measured hypergraph landscapes.

The expected balance cost is

$$\mathbb{E}\big[(\textstyle\sum_i x_i - b)^2\big] = (\textstyle\sum_i p_i - b)^2 + \textstyle\sum_i p_i(1-p_i). \quad (9)$$

Dropping the variance term would change the variational objective. For the 1% and 3% runs, one correlated slack block is independent of the vertex and net blocks. If g_s counts the binary representations of slack value s, conditional minimization gives $q_s \propto g_s \exp[-\lambda_t(\mu - L - s)^2/T]$, where $\mu = \sum_i p_i$. Multiplicity retains the entropy of the underlying binary states. The expected balance term is $\lambda_t[(\mu - L - \mathbb{E}S)^2 + \sum_i p_i(1-p_i) + \mathrm{Var}(S)]$. The current slack expectation provides the per-replica target in the same implicit vertex update, and is refreshed before each update. Binary readout minimizes slack and net auxiliaries with vertices fixed.

FEM represents vertex marginals by sigmoid fields and separates sparse interaction gradients from the global count penalty. The implicit count correction is a scalar root with strictly positive derivative, so it can be solved with a bracketed update. This remains a soft-penalty variational step, without projection onto a balanced binary set. The field equations, preconditioning, clipping and numerical checks are specified in Supplementary Note 4.

Constructing temporary search paths

With the distribution fixed, local interactions and collective directions control the intermediate response. Their envelopes return every temporary term to zero before the original-objective readout.

Local incidence response

For the pair catalyst, the base coefficient is $a_e^{(0)} = 1/\max(1, \lfloor r_e^2/4 \rfloor)$. Node scales take the form $z_i = (1 + D_i)^{-\alpha} \exp(\xi s_i)$, where s_i is a bounded structural signal. Coefficients are rescaled so that $\sum_e w_e a_e \sum_{i<j\in e} z_i z_j$ equals its base value. Degree signals use $\ln(1 + D_i)$; local contrast subtracts the weighted neighbor mean, counting repeated net co-occurrences. Signals are centered, standardized, clipped to two standard deviations, recentered and bounded; isolated vertices receive zero signal. Incidence operations compute these quantities without constructing a clique adjacency matrix.

Collective balanced directions

Let $\mathbf{B}$ be net-by-vertex incidence, $\mathbf{r}$ the net sizes and $a_e = 1/\max(1, \lfloor r_e^2/4 \rfloor)$. The implicit operator is $\mathbf{L} = \mathrm{diag}(\mathbf{B}^\mathsf{T}(\mathbf{a} \odot \mathbf{r})) - \mathbf{B}^\mathsf{T}\mathrm{diag}(\mathbf{a})\mathbf{B}$. Its products use sparse incidence without storing clique edges. Let $\mathbf{D}_c = \mathrm{diag}((L_{ii})_i)$ on the cut-active support, and let $\mathbf{P}_c$ remove the mean separately within each connected component. We also compute modes of

$$\mathbf{L}_{\mathrm{norm}} = \mathbf{P}_c \mathbf{D}_c^{-1/2} \mathbf{L} \mathbf{D}_c^{-1/2} \mathbf{P}_c. \quad (10)$$

These degree-normalized physical directions are not eigenvectors of the unscaled operator. Inactive vertices are excluded only from these eigenproblems and retained in the complete QUBO.

Each geometry supplies four zero-mean, orthonormal physical directions. On connected support we discard the constant mode. On disconnected support, random component contrasts occupy its nonconstant nullspace. If the largest component exceeds half the total vertex count, we keep one contrast and three internal directions; otherwise, up to four contrasts are retained. Internal modes use a shift on the component nullspace. The Lanczos solver uses tolerance 10^{-6}, maximum 20,000 iterations, subspace size 32 and probe seed 314159; absolute residuals are checked below 10^{-5}. Modes are padded with zeros on inactive spins.

For $\mathbf{m} = 2\mathbf{p} - \mathbf{1}$, one catalyst's exact Bernoulli expectation is $-\rho[(\mathbf{v}^\mathsf{T}\mathbf{m})^2 - \sum_i v_i^2 m_i^2]/2$, with gradient $-2\rho[\mathbf{v}(\mathbf{v}^\mathsf{T}\mathbf{m}) - \mathbf{v}^{\odot 2} \odot \mathbf{m}]$. The diagonal is removed because binary $s_i^2 = 1$. Weighted subspaces sum these quadratic expectations. The 48-replica policy divides equally among local, unscaled-mode and normalized-mode groups. All groups receive the same nearby random initial fields, with local, auxiliary and per-spin controls acting simultaneously. Supplementary Note 4 specifies the mode weights, strength grid and component-nullspace schedules.

For $f > 1$ inactive spins, each null direction assigns $1/\lfloor f/2 \rfloor$ to one random subset and $-1/\lceil f/2 \rceil$ to its complement, then normalizes. Thus it has zero mean for odd or even support sizes; one inactive spin cannot supply such a direction. *thermal1* has 248 inactive spins and *ca-CondMat* has 610. Fixed replica-dependent seeds

match the two collective groups. Local replicas have no null coupling.

Schedules and replica allocation

Temporary local temperatures are $T_i^a(t) = T(t)\exp[\omega(t)\zeta_a s_i]$, with $\omega(t) = \sin^2(\pi t/0.9)$ for normalized annealing time $0 \le t < 0.9$ and zero thereafter. The pair and auxiliary envelopes rise over the first 10% of annealing and decay over the final 25%, using squared-sine ramps. Auxiliary strengths multiply $w_e \max(r_e/2 - 1, 0)$. The balance penalty remains 0.01 for the first 90% and increases geometrically to $D_{\max} + 1$ over the final 10%. In the primary benchmark, temperature decreases exponentially from 8 to 0.02 over 12,000 updates; 144 additional updates retain the final temperature, full penalty and zero catalysts. Fixed checkpoints occur every 128 updates and at both phase boundaries. Selection uses the original full-QUBO energy.

Each 16-replica batch starts from a shared normal field with standard deviation 0.01 plus independent perturbations with standard deviation 0.001. Base seeds determine both fields reproducibly. The fixed portfolio combines negative and positive contrast weighting, degree-temperature biases, local-contrast cooling, inverse-degree geometry, a geometry transition and uniform auxiliary alternatives. Supplementary Note 4 gives the profile parameters. Replicas evolve as independent variational starts, without the temperature-swap transitions of exchange Monte Carlo[39]. Per-spin temperature offsets are classical variational controls; they do not implement the inhomogeneous transverse-field schedules studied in quantum annealing[40]. No graph name determines a profile or parameter.

The collective envelope rises over annealing fractions 0–0.15 and falls over 0.80–1; the null envelope rises over 0.55–0.70 and falls over 0.96–1. Squared-sine ramps give zero endpoints. Null strengths multiply $D_{\max} + 1$ by $\{0.5, 1, 2, 4\}$.

Temperature scheduling follows classical annealing[41]. Separate classical references stably sort each of the eight modes and assign exactly $\lfloor n/2 \rfloor$ vertices to one block. These binary partitions never enter FEM.

Local response and mechanism checks

The reduced free energy connects the specified path to a testable local response. We evaluate its balanced-subspace curvature, then check the analytic expressions against independent derivatives and enumeration.

For even n, uniform vertex probabilities are a stationary point of the reduced block-variational objective when the catalyst has no linear field. At uniform temperature, Eq. (4) gives its vertex-probability Hessian, where $\mathbf{A}_{\text{off}}$ contains the off-diagonal pair weights $\sum_{e \ni i,j} w_e a_e z_i z_j$, and

$$\chi_e = \left[1 + \exp(-\kappa_e/T)\cosh\big(w_e(r_e/2 - 1)/T\big)\right]^{-1}. \tag{11}$$

Indeed, $\chi_e = \mathrm{Var}_{q_e}(v_e - u_e)$ at the uniform state; minimizing the auxiliary Gibbs block contributes its negative susceptibility to the Hessian. The entropy Hessian is $4T\mathbf{I}$, and the exact balance expectation contributes $2\lambda(\mathbf{1}\mathbf{1}^{\mathsf{T}} - \mathbf{I})$. The relevant balanced-subspace operator is $\mathbf{PHP}$, with $\mathbf{P} = \mathbf{I} - \mathbf{1}\mathbf{1}^{\mathsf{T}}/n$. Per-spin temperatures replace the entropy term by $4\,\mathrm{diag}(T_i)$. This expression is checked against automatic second derivatives of the complete reduced free energy on weighted small hypergraphs, with and without auxiliary polarization. Matrix-free incidence products evaluate the operator without storing a clique matrix. Its eigenvalues describe local classical free-energy curvature.

Fig. 2(c) uses example B, a penalty of 0.01 and 100 logarithmically spaced temperatures from 0.04 to 8. Auxiliary polarization is zero; exact balanced-subspace eigenvalues are computed from the verified matrix-free Hessian.

Scope of the geometry analysis

For a fixed cut threshold τ, define the feasible low-cost set $\mathcal{S}_\tau = \{\mathbf{x} : \sum_i x_i = b,\ E_{\text{cut}}(\mathbf{x}) \le \tau\}$. The complement-invariant vertex overlap is $q(\mathbf{x}, \mathbf{y}) = |n^{-1}\sum_i (2x_i - 1)(2y_i - 1)|$. An overlap-gap statement requires a forbidden interval of such overlaps between appropriate low-cost states, with its ensemble and threshold specified. The present real-graph runs do not establish that property. Nor does a catalyst automatically evade an algorithmic obstruction: the resulting algorithm may still belong to a class for which a stability-based theorem applies. The curvature analysis concerns local variational stability only.

Partition evaluation and computational cost

The partitioning study uses 30 circuit-netlist, matrix-derived and network hypergraphs from public benchmark collections[42,43], with two blocks and unit loads and net weights. At each allowed imbalance (0%, 1%, 3%), one 48-replica batch receives 12,000 annealing and 144 cold updates, giving 4,320 primary trajectories. The policy is fixed before evaluating each cohort. Initial fields are random and are shared across matched controls; reference and rounded spectral partitions never initialize FEM.

Every replica retains its lowest original-QUBO checkpoint energy at 128-update intervals and phase boundaries. Vertex signs give raw labels; net auxiliaries and slack are minimized with vertices fixed. Original- polynomial one-bit descent then runs on all 48 replicas, without paired moves, binary projection or a feasibility filter on proposed moves. Initial fields and complete raw and selected states are independently reconstructed and scored. Feasibility requires the exact integer capacity and zero auxiliary gap; failed or incomplete outputs are retained. Readout and validation rules are given in Supplementary Note 7.

Computational cost includes mode construction, numerical setup, the complete annealing batch and checkpoints, and descent on all replicas. The primary search uses double precision on a GPU; mode construction is performed on a CPU, with safeguarded updates and a certified mode-construction alternative for *scircuit* specified in Supplementary Notes 4 and 6. Both mode-construction costs are charged even when the jobs overlap.

KaHyPar receives the same original hypergraph, cut objective and integer capacity. Its returned partitions are independently rescored, and measured elapsed time accompanies each result. The ordinary and evolutionary reference configurations are described in Supplementary Note 1; hardware and timing definitions are given in Supplementary Note 6. The comparisons assess quality at matching capacity, while different hardware and time budgets limit speedup interpretations.

Schedule-mixture, penalty-feedback and precision controls address distinct numerical questions and retain their specified budgets (Supplementary Notes 5 and 6). They are evaluated separately from the primary quality and timing results. A single-input precision control cannot establish a general quality effect.

Data availability

Aggregate benchmark results and input provenance are available at https://github.com/yao-baijian/qubo-hypergraph-partition/tree/main/docs/benchmarks.

Code availability

The reference FEM implementation and partitioning interface are available at https://github.com/yao-baijian/qubo-hypergraph-partition.

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